

TASK-1

Subnetting 192.168.10.0/24 into 4 Equal Subnets

Given Network: 192.168.10.0/24

To divide the network into **4 equal subnets**, borrow **2 bits** from the host portion:

- **New CIDR: /26**
- **Subnet Mask: 255.255.255.192**
- **Number of Subnets: 4**
- **Addresses per Subnet: 64**
- **Usable Hosts per Subnet: 62**

Subnet Details

Subnet	Subnet Mask	Network Address	Broadcast Address
Subnet 1	255.255.255.192 (/26)	192.168.10.0	192.168.10.63
Subnet 2	255.255.255.192 (/26)	192.168.10.64	192.168.10.127
Subnet 3	255.255.255.192 (/26)	192.168.10.128	192.168.10.191
Subnet 4	255.255.255.192 (/26)	192.168.10.192	192.168.10.255

Usable Host Ranges

Subnet	Usable Host Range
Subnet 1	192.168.10.1 – 192.168.10.62
Subnet 2	192.168.10.65 – 192.168.10.126
Subnet 3	192.168.10.129 – 192.168.10.190
Subnet 4	192.168.10.193 – 192.168.10.254

Summary

- **Original Network:** 192.168.10.0/24
- **New Subnet Mask:** 255.255.255.192 (/26)
- **Total Subnets:** 4
- **Usable Hosts per Subnet:** 62

TASK-2

Using **Subnet 1** from the previous answer:

- **Network Address:** 192.168.10.0/26
- **Subnet Mask:** 255.255.255.192
- **Usable Host Range:** 192.168.10.1 – 192.168.10.62
- **Broadcast Address:** 192.168.10.63

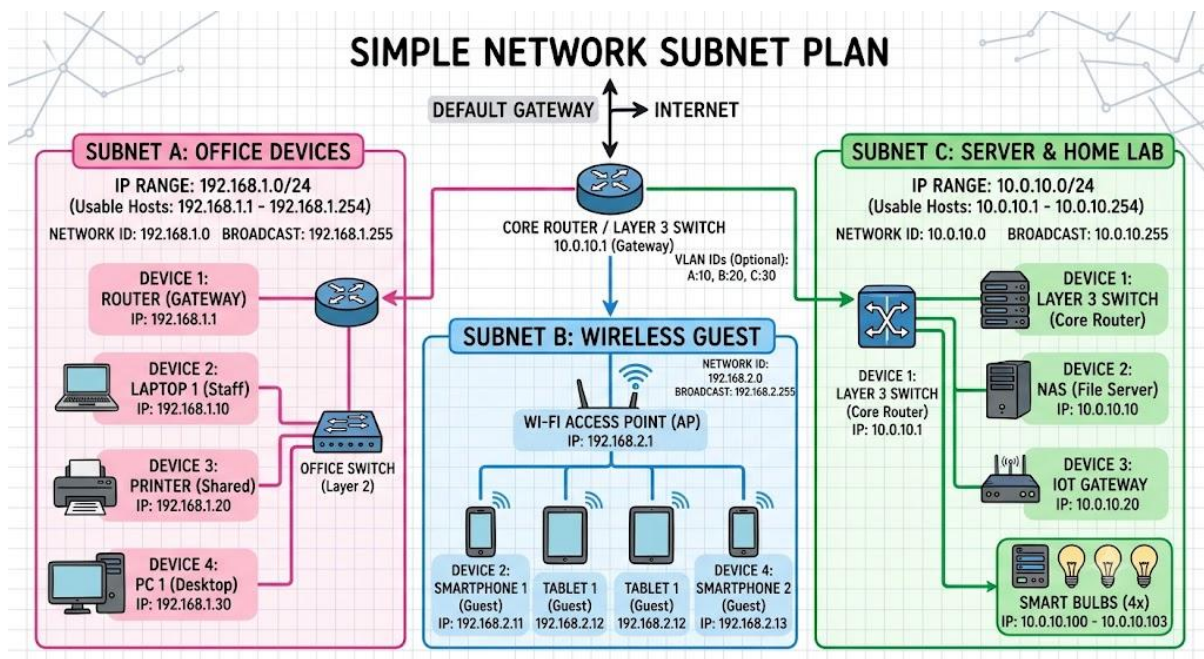
IP Address Assignment

Device	Assigned IP Address	Subnet Mask	Default Gateway
Router	192.168.10.1	255.255.255.192	N/A
Laptop 1	192.168.10.2	255.255.255.192	192.168.10.1
Laptop 2	192.168.10.3	255.255.255.192	192.168.10.1
Printer	192.168.10.4	255.255.255.192	192.168.10.1
Mobile Phone	192.168.10.5	255.255.255.192	192.168.10.1

Note

- **Network Address (192.168.10.0)** is **not assigned** to any device.
- **Broadcast Address (192.168.10.63)** is **not assigned** to any device.
- All assigned IP addresses are **valid host addresses** within the **192.168.10.0/26** subnet.

TASK-3



TASK-4

Step-by-Step Subnetting Explanation (Beginner Friendly)

Step 1: Start with the Original Network

We are given the network:

- **Network Address:** 192.168.10.0/24

The **/24** means:

- The first **24 bits** are used for the network.
- The last **8 bits** are available for host devices.

This gives **256 total IP addresses** (0–255).

Step 2: Decide How Many Subnets We Need

The question asks us to create **4 equal subnets**.

To make 4 subnets, we borrow **2 bits** from the host portion because:

- **$2^2 = 4$ subnets**

So the subnet mask changes from:

- **/24 → /26**

The new subnet mask is:

255.255.255.192

Step 3: Calculate the Number of Hosts per Subnet

After borrowing 2 bits, we have:

- **$32 - 26 = 6$ host bits**

Calculate the addresses:

- **$2^6 = 64$ total addresses per subnet**

Out of these:

- **1 Network Address**
- **1 Broadcast Address**

So:

- **$64 - 2 = 62$ usable host IP addresses**
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Step 4: Find Each Subnet

The subnet size is **64**, so each subnet starts after every 64 addresses.

	Subnet	Network Address	Broadcast Address
1		192.168.10.0	192.168.10.63
2		192.168.10.64	192.168.10.127
3		192.168.10.128	192.168.10.191
4		192.168.10.192	192.168.10.255

Step 5: Choose One Subnet

I selected **Subnet 1**.

- **Network Address:** 192.168.10.0
- **Broadcast Address:** 192.168.10.63
- **Usable IP Range:** 192.168.10.1 – 192.168.10.62

The first and last addresses cannot be assigned to devices because:

- **192.168.10.0** identifies the network.

- **192.168.10.63** is the broadcast address.

Step 6: Assign IP Addresses to Devices

Device	Assigned IP Address
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Router	192.168.10.1
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Laptop 1	192.168.10.2
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Laptop 2	192.168.10.3
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Printer	192.168.10.4
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Mobile Phone	192.168.10.5
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All devices use:

- **Subnet Mask:** 255.255.255.192
- **Default Gateway:** 192.168.10.1 (Router)

Final Summary

Item	Value
Original Network	192.168.10.0/24
Required Subnets	4
New CIDR	/26
Subnet Mask	255.255.255.192
Total Subnets	4
Total Addresses per Subnet	64
Usable Hosts per Subnet	62
Selected Subnet	192.168.10.0/26
Usable Host Range	192.168.10.1 – 192.168.10.62
Broadcast Address	192.168.10.63

Simple Explanation

Think of a **/24 network as one large apartment building with 256 room numbers**. We wanted **4 separate sections**, so we divided the building into **4 equal blocks**. Each block has **64 room numbers**, but the **first room (network address)** identifies the block and the **last room (broadcast address)** is reserved for announcements. That leaves **62 room numbers** that can be assigned to devices like routers, laptops, printers, and phones.